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Jet Impact on Surfaces

Fluid Mechanics — Laboratory Session Report

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1 Introduction

When a liquid jet hits a solid surface, the change in its linear momentum causes a force on that surface. This momentum transfer is one of the ways in which fluid streams produce forces, and it can be calculated without solving for the detailed flow in the impact region: only the fluxes of momentum in the boundaries of a chosen control volume are needed.

The specific objectives of this session are:

1. to derive, from the continuity and momentum equations applied to a control volume, the force exerted by a jet on a flat plate, a 30° oblique surface and a hemispherical surface;
2. to measure that force experimentally by balancing it with calibrated masses for several flow rates;
3. to compare $F(Q)$ with the theoretical predictions and to compute the drag coefficient C_d of each obstacle as a function of the Reynolds number Re_d ;
4. to quantify the measurement uncertainty and to diagnose the physical origin of the discrepancies.

2 Theoretical Derivation of the Jet Impact Force

2.1 Control Volume Formulation and Bernoulli Equation

Consider the control volume bounded by the closed surface $\Sigma_c = \Sigma_i + \Sigma_o + \Sigma_l + \Sigma_w$, where:

- Σ_i : the inlet jet surface arriving to the plate (area A_i).
- Σ_o : the outlet jet surface exiting the water sheet (area A_o).
- Σ_l : the lateral surface of the jet exposed to atmospheric pressure ($p = p_a$).
- Σ_w : the contact surface on the obstacle.

Applying Bernoulli's equation to the streamline from the nozzle exit to the sheet exit and neglecting the effect of gravity:

$$\frac{p_a}{\rho} + \frac{v_i^2}{2} = \frac{p_a}{\rho} + \frac{v_o^2}{2} \implies v_i = v_o = v = \frac{Q}{A} \quad (2.1)$$

2.2 Conservation of Mass (Mass Continuity Equation)

For steady, incompressible flow through the control surface Σ_c :

$$\iint_{\Sigma_c} \rho(\vec{v} \cdot \vec{n}) d\sigma = 0 \quad (2.2)$$

Splitting across the boundary surfaces:

$$\iint_{\Sigma_i} \rho(\vec{v} \cdot \vec{n}) d\sigma + \iint_{\Sigma_o} \rho(\vec{v} \cdot \vec{n}) d\sigma + \iint_{\Sigma_l} \rho(\vec{v} \cdot \vec{n}) d\sigma + \iint_{\Sigma_w} \rho(\vec{v} \cdot \vec{n}) d\sigma = 0 \quad (2.3)$$

Since $\vec{v} \cdot \vec{n} = 0$ in the stream surfaces Σ_l and Σ_w , with $\vec{v}_i = v_i \vec{e}_y$, $\vec{n}_i = -\vec{e}_y$, and $\vec{v}_o \cdot \vec{n}_o = v_o$:

$$-\rho v_i A_i + \rho v_o A_o = 0 \implies \rho v_i A_i = \rho v_o A_o \quad (2.4)$$

Since $v_i = v_o = v$, it follows that:

$$A_i = A_o = A \quad (2.5)$$

2.3 Conservation of Linear Momentum

The conservation of linear momentum equation applied to the control volume Σ_c neglecting the effects of the viscous and gravity forces is:

$$\iint_{\Sigma_c} \rho \vec{v}(\vec{v} \cdot \vec{n}) d\sigma = - \iint_{\Sigma_c} (p - p_a) \vec{n} d\sigma \quad (2.6)$$

Along the different surfaces:

$$- \iint_{\Sigma_c} (p - p_a) \vec{n} d\sigma = - \underbrace{\iint_{\Sigma_i} (p_a - p_a) \vec{n} d\sigma}_0 - \underbrace{\iint_{\Sigma_o} (p_a - p_a) \vec{n} d\sigma}_0 - \underbrace{\iint_{\Sigma_l} (p_a - p_a) \vec{n} d\sigma}_0 - \iint_{\Sigma_w} (p - p_a) \vec{n} d\sigma \quad (2.7)$$

The force exerted by the jet on the obstacle accounts for the atmospheric pressure p_a acting on the upper face of the obstacle:

$$\vec{F} = \iint_{\Sigma_w} (p - p_a) \vec{n} d\sigma \implies - \iint_{\Sigma_c} (p - p_a) \vec{n} d\sigma = -\vec{F} \quad (2.8)$$

Therefore, the momentum equation reduces to:

$$\iint_{\Sigma_i} \rho \vec{v}(\vec{v} \cdot \vec{n}) d\sigma + \iint_{\Sigma_o} \rho \vec{v}(\vec{v} \cdot \vec{n}) d\sigma = -\vec{F} \quad (2.9)$$

2.4 Evaluation of the Force on the Three Geometries

Flat Surface ($\beta = 90^\circ$) At the inlet $\vec{v}_i = v_i \vec{e}_y$ with $\vec{n}_i = -\vec{e}_y$. At the outlet, the fluid leaves purely horizontally along the plate: $\vec{v}_o = v_o \vec{e}_x$ with $\vec{v}_o \cdot \vec{n}_o = v_o$:

$$\int_{\Sigma_i} \rho (v_i \vec{e}_y)(-v_i) d\sigma + \rho v_o^2 A_o \vec{e}_x = -\vec{F} \quad (2.10)$$

$$-\rho v_i^2 A_i \vec{e}_y + \rho v_o^2 A_o \vec{e}_x = -\vec{F} \implies \vec{F} = \rho v_i^2 A_i \vec{e}_y - \rho v_o^2 A_o \vec{e}_x \quad (2.11)$$

Projecting along the vertical direction $\vec{e}_y$:

$$F_y = \rho v_i^2 A_i = \rho v^2 A = \rho \left(\frac{Q}{A} \right)^2 A = \rho \frac{Q^2}{A} \quad (2.12)$$

Oblique Surface ($\alpha = 30^\circ$) The water sheet leaves inclined at an angle of 30° below the horizontal:

$$\vec{v}_o = v_o \cos\left(\frac{\pi}{6}\right) \vec{e}_x - v_o \sin\left(\frac{\pi}{6}\right) \vec{e}_y, \quad \text{with } \vec{v}_o \cdot \vec{n}_o = v_o \quad (2.13)$$

Substituting into the momentum balance:

$$-\rho v_i^2 A_i \vec{e}_y + \int_{\Sigma_o} \rho \left(v_o \cos\left(\frac{\pi}{6}\right) \vec{e}_x - v_o \sin\left(\frac{\pi}{6}\right) \vec{e}_y \right) v_o d\sigma = -\vec{F} \quad (2.14)$$

$$\vec{F} = \rho v_i^2 A_i \vec{e}_y + \rho v_o^2 \cos\left(\frac{\pi}{6}\right) A_o \vec{e}_x + \rho v_o^2 \sin\left(\frac{\pi}{6}\right) A_o \vec{e}_y \quad (2.15)$$

Since $\sin(\pi/6) = \frac{1}{2}$ and $v_i = v_o = v$, $A_i = A_o = A$, the vertical force F_y is:

$$F_y = \left(1 + \sin\left(\frac{\pi}{6}\right) \right) \rho v^2 A = \left(1 + \frac{1}{2} \right) \rho v^2 A = \frac{3}{2} \rho \frac{Q^2}{A} \quad (2.16)$$

Hemispherical Surface ($\beta = 180^\circ$) The water sheet exits directed completely downwards: $\vec{v}_o = -v_o \vec{e}_y$ with $\vec{n}_o = -\vec{e}_y$, so $\vec{v}_o \cdot \vec{n}_o = (-v_o \vec{e}_y) \cdot (-\vec{e}_y) = v_o$:

$$\int_{\Sigma_i} \rho(v_i \vec{e}_y)(-v_i) d\sigma + \int_{\Sigma_o} \rho(-v_o \vec{e}_y)(v_o) d\sigma = -\vec{F} \quad (2.17)$$

$$-\rho v_i^2 A_i \vec{e}_y - \rho v_o^2 A_o \vec{e}_y = -\vec{F} \quad (2.18)$$

$$\vec{F} = 2\rho v^2 A \vec{e}_y \implies F_y = 2\rho v^2 A = 2\rho \frac{Q^2}{A} \quad (2.19)$$

2.5 Drag Coefficient (C_d)

If we define the drag coefficient as the dimensionless number:

$$C_d = \frac{F}{\frac{1}{2}\rho v^2 A} \quad (2.20)$$

substituting the vertical force expressions derived for each geometry:

- **Flat surface** ($F = \rho v^2 A = \rho \frac{Q^2}{A}$):

$$C_d = \frac{\rho v^2 A}{\frac{1}{2}\rho v^2 A} = 2 \quad (2.21)$$

- **Oblique surface** ($F = \frac{3}{2}\rho v^2 A = 1.5\rho \frac{Q^2}{A}$):

$$C_d = \frac{\frac{3}{2}\rho v^2 A}{\frac{1}{2}\rho v^2 A} = 3 \quad (2.22)$$

- **Hemispherical surface** ($F = 2\rho v^2 A = 2\rho \frac{Q^2}{A}$):

$$C_d = \frac{2\rho v^2 A}{\frac{1}{2}\rho v^2 A} = 4 \quad (2.23)$$

Thus, the theoretical drag coefficients take the constant values $C_d = 2$ for the flat surface, $C_d = 3$ for the oblique surface, and $C_d = 4$ for the hemispherical surface.

3 Experimental Apparatus and Test Procedure

3.1 Apparatus

To calculate the linear momentum transfer to the obstacle, a vertical nozzle with an inner diameter of $d = 8$ mm directs the jet toward the obstacle inside a test cylinder. Three interchangeable obstacle geometries were tested: a flat plate (90° radial deflection), a 30° conical one, and a hemispherical cup (180° flow reversal). The experimental flow rate was monitored through the flow meter in L/h. The reaction force is measured by stacking 50 g and 100 g weights on a rod attached to the obstacle. Before each test, the flow was increased in 100 L/h steps. By stacking weight disks on top of the system, the impact force is equilibrated with the weight ($F = mg$).

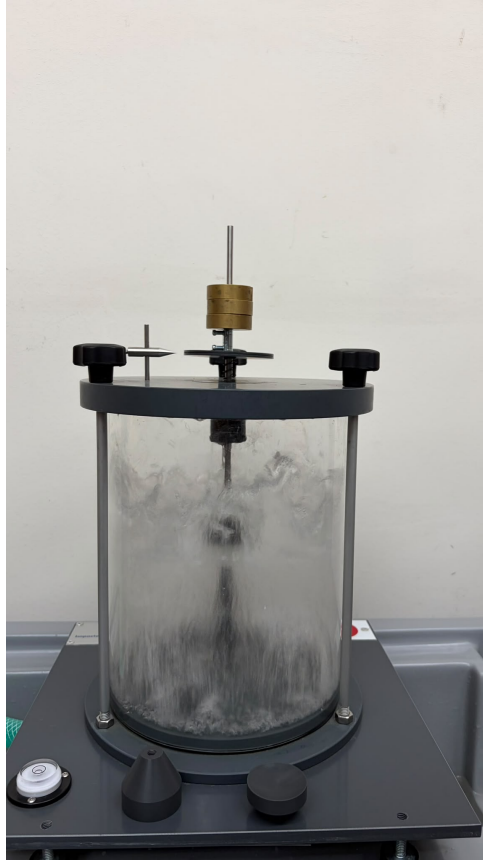


Figure 1: Jet impact experimental setup during flow measurement, balanced with calibrated weights.



(a) First flow condition



(b) Second flow condition

Figure 2: Flowmeter at different flow rates.

3.2 Procedure

1. Remove the cover of the transparent chamber, fix the flat plate to the vertical rod at the impact position and close the chamber.
2. Place the module in the channel of the bench, connect its water inlet to the pump and level the assembly.
3. Adjust the dial gauge until it is at the same height as the mark of the auxiliary platform.
4. Select a fixed flow rate at which the dial gauge stops being at equilibrium (500 L/h in all three parts of our experiment)
5. Place a calibrated mass on the platform, see if the dial gauge reaches again the equilibrium point, if so, note down both values, if not, keep increasing the external load on the disk until it does.
6. Once a measure has been noted down, increase the flow rate again (we did steps of 100 L/h), see if it surpasses the equilibrium point and repeat the loading process from step 5.
7. Repeat the whole procedure with the oblique deflector and the hemispherical cup.

Note: the procedure followed in the lab is not the same as the one presented in the laboratory guidelines, which explains why our results are not the most accurate.

In the data set of this report, we obtained eleven equilibrium points for each obstacle, with flow rates from 1500 L/h to 500 L/h in steps of 100 L/h. The fluid properties used are $\rho = 1000 \text{ kg/m}^3$, $\mu = 1.0 \times 10^{-3} \text{ Pa}\cdot\text{s}$ and $g = 9.81 \text{ m/s}^2$, giving $A = 5.027 \times 10^{-5} \text{ m}^2$.

4 Data Reduction and Uncertainty Calculation

All data least-squares fits and figures were produced with MATLAB. The flow rate is converted to SI units, $Q_{\text{SI}} = Q_{\text{l/h}}/(3.6 \times 10^6)$ (in m^3/s), and then $v = Q/A$, $Re_d = \rho v d / \mu$, $F = W = (m/1000)g$ and C_d from Eq. (2.20). Table 1 lists the kinematic and hydrodynamic variables.

Table 1: Kinematic and flow variables of the jet ($d = 8 \text{ mm}$).

Q (l/h)	Q ($10^{-4} \text{ m}^3/\text{s}$)	v (m/s)	Re_d	$\rho Q^2/A$ (N)
1500	4.1667	8.289	66 315	3.454
1400	3.8889	7.737	61 894	3.009
1300	3.6111	7.184	57 473	2.594
1200	3.3333	6.631	53 052	2.210
1100	3.0556	6.079	48 631	1.857
1000	2.7778	5.526	44 210	1.535
900	2.5000	4.974	39 789	1.243
800	2.2222	4.421	35 368	0.982
700	1.9444	3.868	30 947	0.752
600	1.6667	3.316	26 526	0.553
500	1.3889	2.763	22 105	0.384

Tables 2–4 contain the data of each obstacle: the calibrated mass m , the measured force W , the theoretical force $F_{\text{th}} = \frac{1}{2} C_d^{\text{th}} \rho Q^2 / A$, the experimental drag coefficient with its uncertainty δC_d .

Table 2: Flat plate (theory: $C_d = 2$).

#	Q (l/h)	m (g)	W (N)	F_{th} (N)	C_d	δC_d
1	1500	350	3.433	3.454	1.99	0.21
2	1400	300	2.943	3.009	1.96	0.23
3	1300	250	2.453	2.594	1.89	0.26
4	1200	250	2.453	2.210	2.22	0.31
5	1100	200	1.962	1.857	2.11	0.36
6	1000	200	1.962	1.535	2.56	0.45
7	900	150	1.472	1.243	2.37	0.53
8	800	100	0.981	0.982	2.00	0.62
9	700	100	0.981	0.752	2.61	0.84
10	600	50	0.491	0.553	1.78	1.04
11	500	0	0.000	0.384	–	–

Table 3: 30° oblique deflector (theory: $C_d = 3$).

#	Q (l/h)	m (g)	W (N)	F_{th} (N)	C_d	δC_d
1	1500	400	3.924	5.181	2.27	0.22
2	1400	350	3.433	4.513	2.28	0.24
3	1300	350	3.433	3.891	2.65	0.29
4	1200	300	2.943	3.316	2.66	0.33
5	1100	250	2.453	2.786	2.64	0.38
6	1000	200	1.962	2.303	2.56	0.45
7	900	150	1.472	1.865	2.37	0.53
8	800	150	1.472	1.474	3.00	0.69
9	700	100	0.981	1.128	2.61	0.84
10	600	100	0.981	0.829	3.55	1.18
11	500	50	0.491	0.576	2.56	1.53

Table 4: Hemispherical cup (theory: $C_d = 4$).

#	Q (l/h)	m (g)	W (N)	F_{th} (N)	C_d	δC_d
1	1500	550	5.396	6.908	3.12	0.25
2	1400	500	4.905	6.017	3.26	0.28
3	1300	450	4.415	5.189	3.40	0.32
4	1200	350	3.433	4.421	3.11	0.35
5	1100	300	2.943	3.715	3.17	0.41
6	1000	250	2.453	3.070	3.20	0.48
7	900	200	1.962	2.487	3.16	0.57
8	800	150	1.472	1.965	3.00	0.69
9	700	100	0.981	1.504	2.61	0.84
10	600	100	0.981	1.105	3.55	1.18
11	500	50	0.491	0.768	2.56	1.53

4.1 Uncertainty propagation

The force is $F = mg$, so $\delta F = g \delta m$. The mass is read as the number of 50 g weights that balances the jet, which means that the true force lies within 25 g of the recorded value; we therefore take $\delta m = 25$ g ($\delta F = 0.245$ N) as the nominal value, and $\delta m = 50$ g ($\delta F = 0.491$ N) as a conservative bound. The flow meter reading is assumed to be known within $\delta Q = 25$ L/h.

Taking logarithms of Eq. (2.20), $\ln C_d = \ln 2 + \ln F + \ln A - \ln \rho - 2 \ln Q$, and differentiating,

$$\frac{dC_d}{C_d} = \frac{dF}{F} - 2 \frac{dQ}{Q}. \quad (4.1)$$

The worst-case (linear) propagation, gives

$$\delta C_d = C_d \left(\frac{\delta F}{F} + 2 \frac{\delta Q}{Q} \right) = C_d \left(\frac{\delta m}{m} + 2 \frac{\delta Q}{Q} \right), \quad \frac{\delta \text{Re}_d}{\text{Re}_d} = \frac{\delta Q}{Q}. \quad (4.2)$$

For the flat plate at $Q = 1500$ L/h, $\delta m/m = 25/350 = 7.1\%$ and $2\delta Q/Q = 3.3\%$, so $\delta C_d/C_d = 10.5\%$, i.e. $C_d = 1.99 \pm 0.21$. At $Q = 600$ L/h ($m = 50$ g) the mass term alone is 50% and $\delta C_d = 1.04$ for $C_d = 1.78$. With $\delta m = 50$ g the corresponding values at 1500 L/h would be ± 0.35 . Because δF is constant while $F \propto Q^2$, the relative uncertainty grows rapidly as the flow rate decreases, and the points at low flow rate have little statistical weight. The uncertainties δC_d in Tables 2–4 use the nominal δm .

4.2 Least-squares fits

Since $F = c Q^2$ with $c = \frac{1}{2} C_d \rho / A$, the effective drag coefficient of each obstacle is obtained from a least-squares fit through the origin of F against Q^2 ,

$$c = \frac{\sum_k Q_k^2 F_k}{\sum_k Q_k^4}, \quad C_d^{\text{exp}} = \frac{2A}{\rho} c, \quad (4.3)$$

with the standard error of the slope $s_c = \sqrt{\sum_k (F_k - c Q_k^2)^2 / [(n-1) \sum_k Q_k^4]}$.

Table 5: Least-squares slopes of $F = c Q^2$ (units of c : 10^7 N s²/m⁶) and effective drag coefficients.

Surface	c_{th}	c_{exp}	C_d^{exp}
Flat plate ($C_d = 2$)	1.989	2.038	2.05 ± 0.07
30° deflector ($C_d = 3$)	2.984	2.448	2.46 ± 0.07
Hemisphere ($C_d = 4$)	3.979	3.180	3.20 ± 0.04

5 Results and Discussion

Figure 3(a–c) shows the measured force against the flow rate together with the theoretical curves for the three obstacles, and Fig. 3(d–f) gives C_d as a function of Re_d . Figure 4 compares the experimental and theoretical data of F as a function of Q for the three tested surfaces. All figures were generated with MATLAB from the data of Tables 2–4.

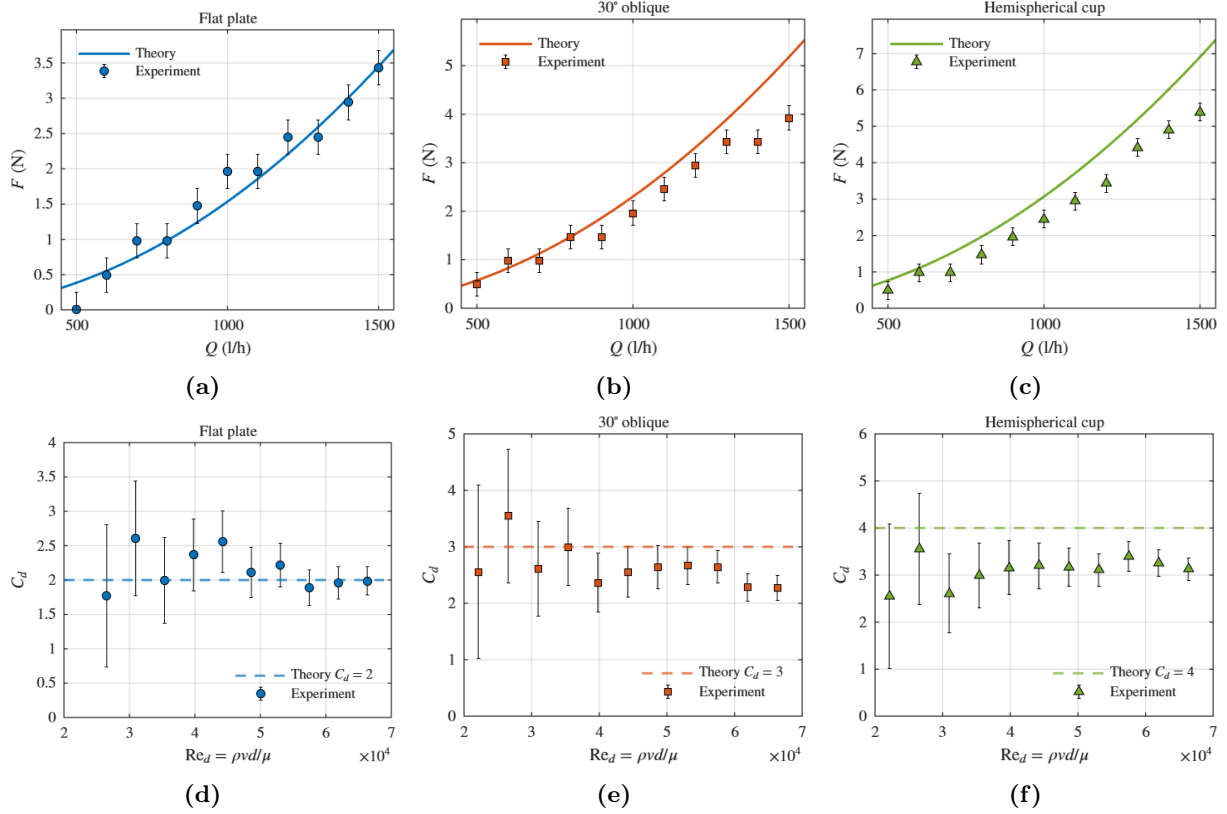


Figure 3: Jet force on the three obstacles: flat plate, 30° oblique plate and hemispherical cup (left to right). (a–c) Force F versus flow rate Q . (d–f) Drag coefficient C_D versus Reynolds number Re . Horizontal error bars are present but not visible.

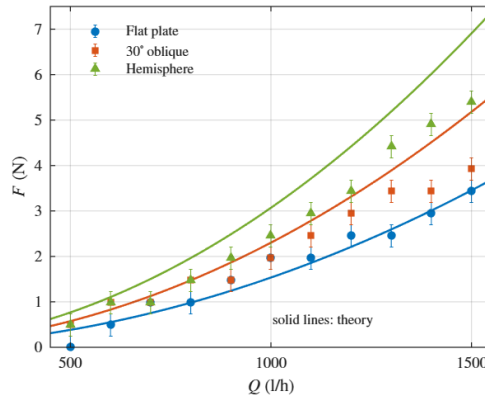


Figure 4: Comparison of the experimental and theoretical data of F as a function of Q for the three tested surfaces.

Flat plate. The experimental slope is $m_{\text{exp}} = 2.038 \times 10^7$ against the theoretical $m = 1.989 \times 10^7$, i.e. $C_d^{\text{exp}} = 2.05 \pm 0.07$ versus $C_d = 2$. At the two highest flow rates the force is within -0.6% and -2.2% of the prediction, and nine of the ten points are compatible with $C_d = 2$ within the error range δC_d . At lower flow rates, the individual errors reach $\pm 30\%$, but they alternate in sign and are of the magnitude expected from the 50 g mass resolution (a single weight is 0.49 N, very close to F_{th} at $Q \leq 700$ L/h). Therefore, we can conclude that the flat plate follows Eq. (2.12) within the experimental uncertainty.

Oblique deflector. The measured slope is 82% of the theoretical one, i.e. $C_d^{\text{exp}} = 2.46 \pm 0.07$ against $C_d = 3$. The deficit is clear at high flow rate, where the measured force is 24% below

the prediction at 1500 and 1400 L/h and 11–15% below between 1300 and 1000 L/h. Six of the eleven points are compatible with $C_d = 3$ within δC_d ; most of them lie at $Q \leq 900$ L/h, where δC_d exceeds 0.5 and the test is therefore weak.

Hemispherical cup. The slope is 80% of the theoretical one, $C_d^{\text{exp}} = 3.20 \pm 0.04$ against $C_d = 4$. This is the best-defined fit and the deficit, -15% to -25% over most of the range, is far larger than the experimental uncertainty. Only two points (those at 600 and 500 L/h, with $\delta C_d > 1$) are compatible with $C_d = 4$.

5.1 Hierarchy of forces

The theory predicts that, with the same rate flow, the forces experimented by each obstacle geometry should follow that $F_{\text{flat}} < F_{30^\circ} < F_{\text{hemi}}$ because the coefficients multiplying the constant $\rho \frac{Q^2}{A}$ are $1 < 1.5 < 2$ respectively. In the experimental data, $F_{\text{flat}} < F_{30^\circ} < F_{\text{hemi}}$ holds strictly at the five highest flow rates (1500 to 1100 L/h); at lower flow rates several readings coincide because of the 50 g resolution (e.g. flat plate and oblique deflector both read 200 g at 1000 L/h, and all three read 100 g at 700 L/h), but the ordering $F_{\text{flat}} \leq F_{30^\circ} \leq F_{\text{hemi}}$ is never violated. The fitted slopes also follow this order strictly ($2.05 < 2.46 < 3.20$), although their ratios, $1 < 1.20 < 1.56$, are smaller than the ideal $1 < 1.5 < 2$.

5.2 Reynolds-number dependence

In Fig. 3(d–f) the flat-plate points scatter around $C_d = 2$. For the deflecting surfaces C_d is below the ideal value throughout, with larger scatter at low Re_d (low force) caused by the low mass sensibility. Over the range of Re_d covered ($2.2\text{--}6.6 \times 10^4$) the data is approximately aligned (as the theory predicts), although its values are aligned below their theoretical values.

6 Sources of Error

The forces measured on the oblique deflector and in the hemisphere are below the ideal ones, which means $C_d^{\text{exp}} < C_d^{\text{th}}$. The following points are the reasons why these discrepancies appear:

1. **Human error.** We consider this as the main source of error due to the incapability of determining with precision the different exact measurements. For example, determining whether the system was balanced and aligned with its equilibrium value or not was hard, as the difference between equilibrium and non-equilibrium was barely millimetric. Also, as 10 people were carrying the experiment at the same time, it was hard to agree with the same data results.
2. **Equipment limitations.** The flow rate values were not completely constant: there was a subtle variation in the flow rate, decreasing and increasing periodically.
3. **Gravitational deceleration of the jet.** The jet travels vertically before it reaches the obstacle, so the impact velocity is lower than the predicted having neglected the effect of gravity in Bernoulli's equation. So, the force caused by the jet impact is actually smaller than the theoretical one we have calculated. This effect explains only a small part of the 20% deficit at high flow rate (it would also affect the flat plate, which agrees with theory), and increases the deviation mainly at low flow rate because the change of the force is relatively bigger at smaller flow rates.

4. **Quantisation of the weights.** The masses are added step by step adding 50 g each time, what corresponds to $\delta W = 0.4905 \text{ N}$. Theoretically, at low flow rates the force itself should be of the order of one or two weights, so then, the relative resolution obtained at the laboratory is poor: the flat plate at 500 L/h ($F_{\text{th}} = 0.384 \text{ N}$, less than one weight) registered 0 g. This is consistent with the force not being able to overcome static friction of the system; in either case the zero reading is a resolution limit and not evidence against the model. This can explain the large $\pm 30\%$ scatter of individual errors at low flow.
5. **Viscous effects.** In a real flow, the water sheet loses some speed because of friction with the walls of the obstacle. This would cause the exit velocity to be lower than the theoretical one. Friction therefore plays a role, together with the previous items, but it cannot explain the whole difference by itself.
6. **Deviation from the prescribed measurement procedure.** The laboratory guide prescribes fixing the mass on the platform and regulating the flow rate until equilibrium, so that the flow meter, which can be read continuously, supplies the measured quantity at each point and the weight $W = mg$ is known exactly. Due to a team's misunderstanding of the laboratory instructions during the session, the opposite was done: the flow rate was fixed at an exact value and the masses were added until the platform was balanced. In principle, both procedures sample the same curve $F(Q)$, but the precision of our method is lower than that of the method in the lab instructions.

7 Conclusions

- Describing the experiment applying the equations we have studied and using approximations in order to be able to solve the equations in the chosen control volume and surface and using the definition of the drag coefficient has resulted in getting theoretical drag coefficients. $C_d = 2, 3, 4$ for the flat plate, the 30° deflector and the hemisphere respectively.
- The measured forces are directly proportional to Q^2 for the three obstacles, and the order of the forces, flat plate $\leq$ oblique $\leq$ hemisphere, is the same at all flow rates.
- For the flat plate the theoretically obtained result is confirmed: $C_d^{\text{exp}} = 2.05 \pm 0.07$, with the force within 2.2% of the prediction at 1400 L/h and above.
- For the 30° deflector and the hemisphere the measured drag coefficients, 2.46 ± 0.07 and 3.20 ± 0.04 , are about 18–20% below the ideal values. The ideal model is therefore an upper bound for these. The reason why our experimental results may be lower than the expected ones are listed in the Sources of Error section.
- The mass resolution of 50 g limits the accuracy at low flow rate, where the relative uncertainty of C_d exceeds 50%. A wider set of weights with smaller load steps, or a longer measurement at high flow rates, would be the most effective improvement of the experiment.
- Drag coefficient C_d as a function of Reynold's number Re_d tends to remain constant as predicted by theory, although our results for the oblique and hemispherical obstacles are below the predicted value.

References

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Tools used

- MATLAB: data reduction, least-squares fits and all figures (`process_lab1_data.m`).
- Overleaf: preparation of the document.